

Here are the detailed, standalone notes for the video on False Discovery Rates (FDR).

Video: False Discovery Rates, FDR, clearly explained

Link: <https://youtu.be/K8LQSVtjcEo>

1. The Core Problem: The Multiple Testing Problem

This video addresses a critical issue in modern statistics (like genomics or RNA-sequencing) where you are **not just running one test, but thousands** at the same time.

- **The Single-Test Fallacy:** When you do *one* statistical test (e.g., comparing Drug A to Control), you set a significance level, typically $\alpha = 0.05$. This means you accept a 5% chance of a **false positive** (a p -value < 0.05 just by random chance, even if the drug does nothing) [03:53].
- **The Multiple-Test Problem:** What happens when you test 10,000 genes at once to see if they are affected by a drug? [04:43]
 - If you apply the $p < 0.05$ rule to all 10,000 genes, you will get a huge number of false positives.
 - **Calculation:** 5% of 10,000 = 500 [04:52].
 - This means you should *expect 500 genes* to appear "statistically significant" ($p < 0.05$) *purely by random chance*, even if the drug has zero effect on any of them.
- **The Goal:** We need a method to "weed out bad data that looks good" [00:46]. This is what the **False Discovery Rate (FDR)** is for.

2. The Shape of p-values: The "Eyeball Method"

To understand how FDR works, you must first understand what a histogram of p-values looks like.

- **Scenario 1: All Tests are "Boring" (All H_0 are True)**
 - If you run 10,000 tests on genes that are *not* affected by the drug, the p-values will be **uniformly distributed**.
 - This means you'll get roughly the same number of p-values in every "bin" (e.g., 500 p-values between 0 and 0.05, 500 between 0.05 and 0.10, etc.) [06:42, 07:15].
 - The histogram looks **flat**.
- **Scenario 2: All Tests are "Interesting" (All H_A are True)**
 - If you run 10,000 tests on genes that *are* affected by the drug, most of your tests should correctly identify this.
 - You will get a huge number of very small p-values (e.g., $p < 0.05$).
 - The histogram looks **heavily skewed to the right**, with a large bar at the far left (near 0) [09:10].
- **Scenario 3: The Real-World Mixture**
 - In a real experiment (e.g., 10,000 genes tested, 1,000 are *really* affected, 9,000 are *not*), your p-value histogram is a **mixture of both shapes** [10:53].
 - It will have a **flat "base"** (from the 9,000 unaffected genes) and a **large "skyscraper"** on the far left (from the 1,000 affected genes).
 - **Crucial Insight:** The p-values on the far left (e.g., in the 0 — 0.05 bin) are a *mix* of **true positives** (the skyscraper) and **false positives** (the flat base) [11:03].
- **The "Eyeball Method":** You can literally "by eye" see the height of the flat, uniform part of the histogram (the false positives) and draw a line across [11:18]. This tells you, roughly, how many of your "significant" results are just part of that flat, random distribution.

3. The Solution: The Benjamini-Hochberg (BH) Method

The **Benjamini-Hochberg (BH) method** is the mathematical formula that performs this "eyeball test" to control the False Discovery Rate [12:38].

- **What it does:** It takes your list of p-values and generates a new, "adjusted" list. This is often called the **"q-value"** or "FDR-adjusted p-value."
- **The Process:**
 - i. Take all your p-values (e.g., 10,000 of them).
 - ii. Order them from **smallest to largest** [15:05].
 - iii. Give them a rank (1, 2, 3, ... 10,000).
 - iv. Apply a formula that "adjusts" each p-value based on its rank and the total number of tests.

- **The Formula (Simplified):** The adjustment formula makes p-values larger.
 - Adjusted p-value = (Original p-value) * (Total # of tests) / (p-value's rank)
 - There's one extra rule: the adjusted p-value can *never be smaller* than the adjusted p-value of the rank below it. This is why the adjustment "crawls" up from the largest p-value [15:40].
- **The Result:**
 - For a truly "boring" p-value (e.g., $p = 0.81$), the adjustment will make it even larger (e.g., $q = 0.90$) [16:10].
 - For a "false positive" p-value (e.g., $p = 0.04$), the adjustment will make it *larger* and non-significant (e.g., $q = 0.06$) [13:11, 16:57].
 - For a truly "interesting" p-value (e.g., $p = 0.0001$), the adjustment will be very small, and it will *remain significant* (e.g., $q = 0.001$).

4. What Does an FDR Cutoff Mean?

This is the most important takeaway.

- If you use a **p-value cutoff of 0.05**, you are saying, "I am willing to accept that 5% of my *uninteresting* tests will be false positives." (5% of 9,000 = 450 false positives).
- If you use an **FDR-adjusted p-value cutoff of 0.05**, you are saying, "I am willing to accept that 5% of the results I call 'significant' are false positives." [13:35]
- **Example:**
 - Your adjusted list (q-values) gives you 400 "significant" genes ($q < 0.05$).
 - This means you *expect* about 5% of those 400 genes (i.e., **20 genes**) to be false positives [13:55].
 - This is *much* better than the 500+ false positives you would have gotten without FDR!

In short: FDR is a correction for multiple testing. It adjusts your p-values to drastically reduce the number of false positives you get when you run thousands of tests at once.

http://googleusercontent.com/youtube_content/9